

Purely Unrectifiable Sets in Metric Spaces

Classical Theory and Recent Developments

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Definitions

Let (X, d) be a (complete) metric space and $k \in \mathbb{N}$.

Definition

A set $S \subset X$ is said to be (countably) k -rectifiable if there exist countable Lipschitz maps $\varphi_n: \mathbb{R}^k \supset A_n \rightarrow X$ such that

$$\mathcal{H}^k(S \setminus \bigcup_{n \in \mathbb{N}} \varphi_n(A_n)) = 0.$$

Definition

A set $S \subset X$ is said to be (countably) purely k -unrectifiable (k -PUN) if

$$\mathcal{H}^k(S \cap \varphi(A)) = 0$$

for every Lipschitz map $\varphi: \mathbb{R}^k \supset A \rightarrow X$.

Examples of (un)rectifiable sets

Every k -submanifold of $\mathbb{R}^n$ is clearly k -rectifiable.

The product of two $\frac{1}{2}$ -dimensional Cantor sets is 1-PUN.

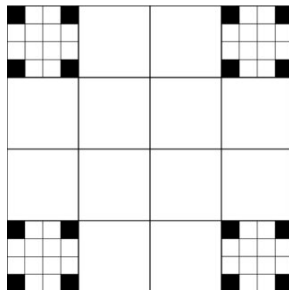


Figure: Second iteration of the 4-corner Cantor set

The classical case

Federer's Structure Theorem

A set $S \subset \mathbb{R}^n$ with $\mathcal{H}^k(S) < \infty$ is k -PUN if and only if

$$\mathcal{H}^k(\pi_K(S)) = 0$$

for a.e. $K \in G(n, k)$, where π_K is the orthogonal projection on K .

It's easy to prove that almost all the projections of a rectifiable set have positive measure. So we just need to prove the following.

If a set S has positive measure projections with positive probability, then there exists a C^1 k -submanifold M such that $\mathcal{H}^k(S \cap M) > 0$.

The proof by B. White for $k = 1$

The proof works by induction on n starting from 2, because the case $(k, n) = (1, 2)$ is well-known since the works of Besicovitch. There are two main steps:

- 1 For any $k < n$ there exists a positive measure subset $\mathcal{S} \subset G(n, k)$ such that for all $V \in \mathcal{S}$ there exists a C^1 curve $C_V \subset V$ such that $\mathcal{H}^1(C_V \cap \pi_V(S)) > 0$.
- 2 Consequently there exist an hyperplane $H \supset C_H$ and a 2-plane $V \supset C_V$ such that the line $V \cap H = L$ is nice for C_H . Then $\pi_H^{-1}(C_H) \cap \pi_V^{-1}(C_V)$ intersects S positively and it's a C^1 curve, being (almost) a transversal intersection.

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The proof by B. White for general k

We'd like to create a C^1 graph that intersects S positively.

Lemma

Let $S \subset \mathbb{R}^n$ be compact with $\mathcal{H}^k(S) < \infty$, $\pi_k: \mathbb{R}^n \rightarrow \mathbb{R}^k$ the standard projection and for a $\rho \in O(n)$ let

$$f_\rho: \pi_k(\rho(S)) \rightarrow \mathbb{R}^{n-k}$$
$$f_\rho(x) = \min_{\text{lex}} \{y \in \mathbb{R}^{n-k} : (x, y) \in \rho(S)\}.$$

Then f_ρ is approximately differentiable for almost every $\rho \in O(n)$.

The proof relies on the Structure Theorem for $k = 1$ and the choice of f_ρ is just a clever way to construct a Borel inverse of an orthogonal projection.

The proof by B. White for general k

Finally

- We can find a $\rho \in O(n)$ such that $\pi_k(\rho(S))$, the domain of f_ρ , has positive $\mathcal{H}^k$ measure.
- Up to restricting the domain, we can assume that f_ρ is Lipschitz and in turn also C^1 by Whitney's Theorem.
- Then its graph, a piece of a C^1 k -submanifold, intersects S positively.

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What is true in general?

Let $m \geq k$ be positive integers.

D. Bate

Let $S \subset X$ with $\mathcal{H}^k(S) < \infty$ be k -PUN and

$$0 < \liminf_{r \rightarrow 0} \frac{\mathcal{H}^k(S \cap B_r(x))}{\omega_k r^k} \leq \limsup_{r \rightarrow 0} \frac{\mathcal{H}^k(S \cap B_r(x))}{\omega_k r^k} < \infty \quad (*)$$

for $\mathcal{H}^k$ -a.e. $x \in S$, then the set

$$\{f \in \text{Lip}_1(X, \mathbb{R}^m) : \mathcal{H}^k(f(S)) = 0\}$$

is residual.

A residual set is a countable intersection of dense open sets, in particular it's dense by Baire's Theorem.

Alberti representations

We'll need a few definitions:

- Let (X, d, μ) be a (complete) metric measure space.
- Let $\Gamma(X)$ be the set of bi-Lipschitz fragments of curves $\gamma: K \rightarrow X$, where $K \subset \mathbb{R}$ is any compact set.

Definition

An Alberti representation (AR) is a measure $\mathbb{P}$ on $\Gamma(X)$ together with measures $\mu_\gamma \ll \mathcal{H}^1 \llcorner \text{Im}(\gamma)$ for all $\gamma \in \Gamma(X)$, such that

$$\mu(E) = \int_{\Gamma(X)} \mu_\gamma(E) d\mathbb{P}(\gamma)$$

for all $E \subset X$.

Weak tangent fields

From now on, let $F: X \rightarrow \mathbb{R}^m$ be a Lipschitz map and $m \geq k$ be positive integers.

Definition

We say that $S \subset X$ admits a k -dimensional weak tangent field with respect to F if there exists a map $\tau: S \rightarrow G(m, k)$ such that for any $\gamma \in \Gamma(X)$ we have $(F \circ \gamma)'(t) \in \tau(\gamma(t))$ for $\mathcal{H}^1$ -a.e. t such that $\gamma(t) \in S$.

We call $\tilde{A}(F, k) \subset \mathcal{P}(X)$ the collection of such sets.

Independent Alberti representations

We call cones the sets of the form

$$C(\theta, w) = \{v \in \mathbb{R}^m : v \cdot w \geq (1 - \theta) \|v\|\}$$

with $w \in \mathbb{S}^{k-1}$ and $\theta \in (0, 1)$.

Definition

An AR is said to be in the F -direction of a cone C if $\mathbb{P}$ -a.e. $\gamma \in \Gamma(X)$ is in the F -direction of C , i.e. $(F \circ \gamma)'(t) \in C \setminus \{0\}$ for $\mathcal{H}^1$ -a.e. $t \in \text{Dom}(\gamma)$.

Definition

A collection of ARs is said to be F -independent if the members are in the F -directions of independent cones.

Connection between ARs and weak tangent fields

Lemma

Every (X, d, μ) admits a decomposition $X = S \cup \bigcup_{n \in \mathbb{N}} U_n$ such that each $\mu \llcorner U_n$ admits k F -independent ARs and $S \in \tilde{A}(F, k-1)$.

The process to construct suitable ARs is divided in some steps:

- 1 By a powerful functional analytic tool, given a cone $C(\theta, w)$ we can decompose $X = S \cup U$ where
 - $\mu \llcorner U$ admits an AR in the F -direction of $C(\theta, w)$.
 - S looks almost F -orthogonal to w .
- 2 Then we ensure that new ARs are F -independent from the previous ones.
- 3 Finally we upgrade sets S that locally look almost F -flat, to sets in $\tilde{A}(F, k-1)$.

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Connections with rectifiability

D. Bate, S. Li

If $S \subset X$ is such that $\mu \llcorner S$ admits k F -independent ARs and $(*)$, then S is (almost) k -rectifiable.

As a simple consequence, we discover that PUN sets look at most $(k - 1)$ -dimensional, from a differentiable point of view.

Corollary

If $S \subset X$ is k -PUN with $\mathcal{H}^k(S) < \infty$ and $(*)$, then

$$S \in \tilde{A}(F, k - 1).$$

for any Lipschitz map $F: X \rightarrow \mathbb{R}^m$.

Perturbations of real-valued Lipschitz maps

Lemma

Let $W \leq \mathbb{R}^m$, suppose that a compact $S \subset X$ looks almost F -parallel to W and that $T: \mathbb{R}^m \rightarrow \mathbb{R}$ is linear with $W \leq \ker(T)$. Then for any $\varepsilon > 0$ there exists a Lipschitz map $f: X \rightarrow \mathbb{R}$ such that

- $\text{Lip}(f) \leq \text{Lip}(T \circ F) + \varepsilon$.
- $\|T \circ F - f\|_\infty \leq \varepsilon$.
- $\text{Lip}(f) \leq \varepsilon$ near to S .

Idea of the proof

The proof is divided in two big steps:

- 1 If X is a Banach space and F is linear, then S is almost flat with respect to $\ker(F)$. Using a cleverly defined functional, we can find f similarly to the usual regularization methods.
- 2 Assuming that F is 1-Lipschitz, we can embed isometrically

$$(F, \iota): X \hookrightarrow \mathbb{R} \times C_b(X)$$

where ι is the Kuratowski embedding. In this way the hypothesis translate to linear ones and we can apply the first step.

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Vectorial perturbations

Let $F: X \rightarrow \mathbb{R}^m$ be a Lipschitz map and $d \in \mathbb{N}$, $s \in \mathbb{R}$ with $s > d$.

Theorem

If $S \in \tilde{A}(F, d)$ with $\mathcal{H}^s(S) < \infty$, then for any $\varepsilon > 0$ there exists a map $\sigma: X \rightarrow \mathbb{R}^m$ such that

- $\text{Lip}(\sigma) \leq \text{Lip}(F) + \varepsilon$.
- $\|\sigma - F\|_\infty \leq \varepsilon$ for all $x \in X$.
- $\mathcal{H}^s(\sigma(S)) \leq \varepsilon$.

Idea of the proof

The proof consists of three main steps, that exploit the intuition that S looks locally flat, but of dimension d lower than s :

- 1 If S looks flat, we extend the real approximation to the vectorial case (component-wise), so that close to S the new σ behaves like a projection on a d -dimensional subspace.
- 2 Orthogonal projections lower the dimensions, so we expect our σ to lower the $\mathcal{H}^s$ -measure, which it does.
- 3 Finally we partition S in flat sets and patch the approximations together, so that σ lowers the measures locally, and in turn globally.

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Typical Lipschitz maps

Theorem

If $S \subset X$ with $\mathcal{H}^s(S) < \infty$ is such that $S \in \tilde{A}(f, d)$ for any Lipschitz map $f: X \rightarrow \mathbb{R}^m$, then

$$\{f \in \text{Lip}_1(X, \mathbb{R}^m) : \mathcal{H}^s(f(S)) = 0\}$$

is residual.

Together with the previous corollary, this results yields the desired conclusion putting $(s, d) = (k, k - 1)$.

Idea of proof

The idea is that, for S compact and $\varepsilon > 0$, the open sets

$$R_\varepsilon(S) = \{f \in \text{Lip}_1(X; \mathbb{R}^k) : f(S) \subset \bigcup_{i=1}^l B(x_i, r_i)\}$$

for some $(x_i)_{i=1}^l$ and $\sum_{i=1}^l r_i^s < \varepsilon$ are also dense. In fact we can deformate a function g into σ so that $\mathcal{H}^s(\sigma(S)) < \varepsilon$, in particular $\sigma(S)$ admits a small covering of balls, i.e. $\sigma \in R_\varepsilon(S)$.

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